



ÉCOLE SUPÉRIEURE DES INDUSTRIES DU TEXTILE ET DE L'HABILLEMENT

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IMS2

QUALITY AGENT

*Real-Time Automated Visual Inspection System Prototype
for Textile Manufacturing Using Multiple Filters*

Technical Report

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Part 1

Introduction

Morocco represents one of the pillars of the textile and garment industry, since its quality consistency pushes into competitive differentiator in the international market. Seeing that with the increase of the global buyers that imposes strict quality standards, manufacturers face today a pressure to achieve higher coverage without proportional increase in labor and harbor.

Our project detects broken yarns, holes, stains, weaving errors and pattern irregularities, these are the major sources of the financial loss we remark through the production chain. Studies estimate that the fabric defects are the main reason for about 45 to 65 percent of incurred losses in fabric sale prices, and that about 85 percent of the product rejections in the industry are the fruits of fabric level defects.

Our objective is to create a system that actually detects automatically the defects with the visual output and the conformity of each piece, to generate automatically quality report and inspections, a follow through report is sent with out email service and what is awesome about this project is that it works in real time.

Project Context

Quality agent comports 5 main steps and is built on 2 stages where we separate detection from diagnosis, this elaborates a fast, specialised and it helped us filter every incoming picture:

- 1- First we capture with either the live webcam feed or from uploading a file.
- 2- Preprocessing, where the image gets resized to a width of 640 pixels, but not only that we also enhance the contrast and its sharpness in order to convert it to the right color expected from our model.
- 3- Detection, the LandingLens supervised model helps us detect the defection with a confidence score. If that score is below 0.75, the decision is escalated to the second filter Mistral as fallback.
- 4- Generative analysis, the Mistral VLM analyses the image then generates a structured report.
- 5- Restitution, a dashboard in Streamlit is our way to visualize, and an alert e-mail is sent for each non-conforming piece.

For the stack, we have Python as our programming language, openCV for image acquisition and the LandingLens SDK from the detection stage, a Mistral API for the analysis, and as we said streamlit for visualization.

Dataset and Model Preparation:

For our dataset, we got it accurate from Kaggle, it is the "Fabric Defect Dataset", that we split into 80% for the training and 20% for the test and validation, the images were labelled defective from non-defective fabric, to detect large categories from stains, seam, stitches and holes using bounding boxes.

Pre-processing Module :

firstly, the image passes through a common pre-processing module (processor.py). this step is applied both to the real-time webcam and to the uploaded images. then it performs three operations :

- Resizing
- contrast enhancement
- sharpness enhancement

it also converts from OpenCV's native BRG colour ordering to RGB, which the inference models expect. without this detail, the quality decreases.

Stage One – Supervised Detection (LandingLens) :

The Only goal of this stage is to detect whether the fabric is defective or not. it is a supervised vision model, fine-tuned on the project's textile images and deployed as a cloud endpoint through landingLens. for each image uploaded, the model returns a label together with a confidence score. it also has a decision threshold of 0.75 meaning the model only predicts "positive" if it is at least 75% confident and no further model is called. When the confidence is below 75% the verdict is not taken from this module alone but hended to the second stage as a fallback.

Stage Two – Decision Module and Verdict Fusion (Mistral fusion with LandingLens) :

The QualityAgent decision use two complementary vision models combined by the score of confidence. The first stage filtre model is the Landing Lens fine-tuned model which generates a verdict of conforming or defective along with a confidence score. when the confidence score is at least 75%, the decision module accepts that verdict, eliminating the need for additional filtre. Otherwise if the confidence score is below the 75% threshold the second stage filtre model (Mistral) will be referenced using a zero-shot inference for our prototypage project. If both verdicts are produced at this stage, the highest confidence score out of the two will be selected as the final output. Once a defect verdict has benn issued, the defect type and also the causes and actions to take will always be identified by Mistral even if it was not collected during the stage of decision module. We fixed the temperature of our VLM to 0.0 so that the same image gives us the same response everytime. Calling the VLM only when a defect has been detected or the Landing Lens confidence score is below 0.75 is to reduce API call, latency and cost.

Real-time Inspection :

the system can operate continuously on a live camera feed. A capture loop reads frames from the webcam through OpenCV and runs an inference once per second rather than 30 per second which costs more, increases latency and is overall non practical. we have the verdict (OK or defect) is overlaid on the video stream using the color green for OK and red for DEFECT.

Interactive Dashboard and Alerting :

All user interaction takes place through a dashboard built with streamlit (app.py). the dashboard offers two mode : a live-camera inspector and uploading images. for each analysis, we have the verdict card and the confidence score, the AI-generated diagnosis, a quality score and the report of the defect. it also has a history of all past analyses providing a traceability. finally, when a defect is

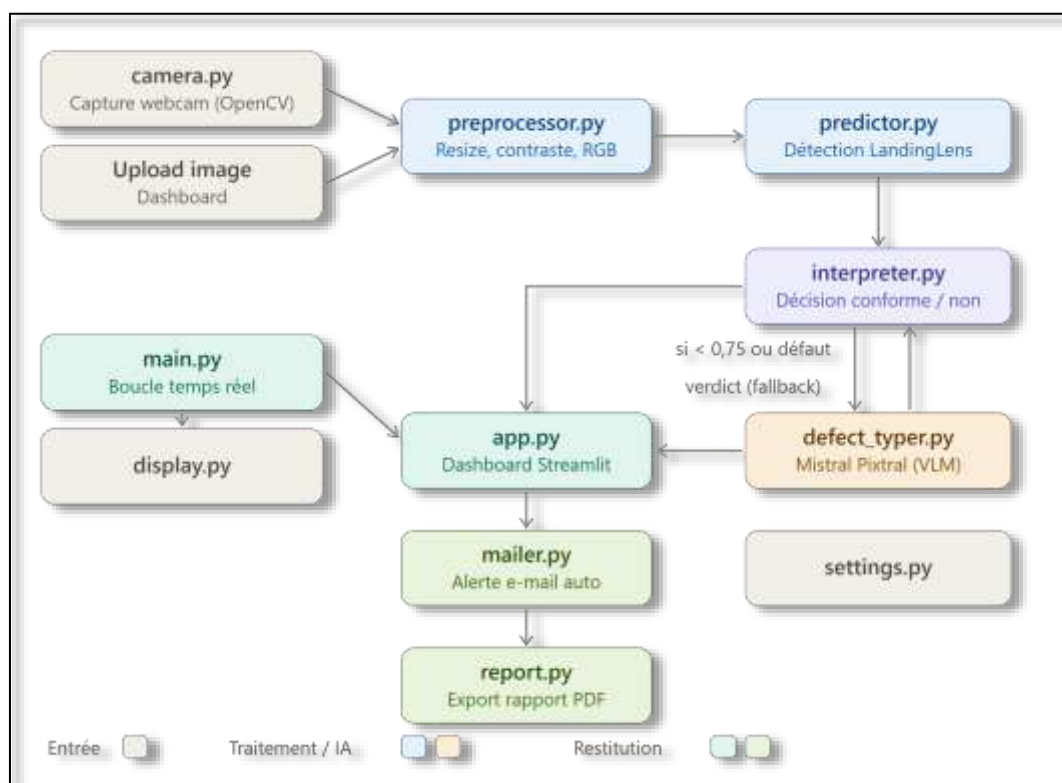
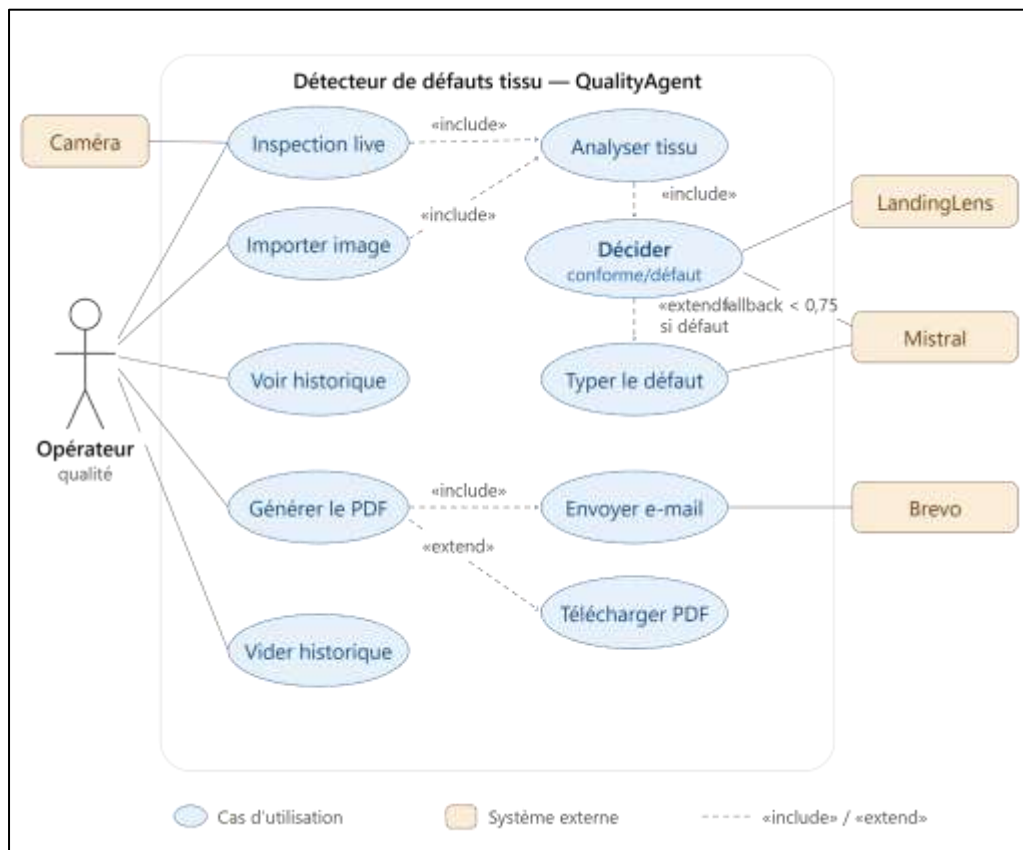
detected, the system automatically send an e-mail alert containing the defect type and all informations related to the non-conformity.

Software Architecture and Modularity :

In our project, each file has its own responsibility. this separation improves maintainability, testability and overall, it makes our project more readable, understandable and organised.

Module	Responsibility
camera.py	Webcam capture through OpenCV, exposed as a safe context manager that always releases the device.
preprocessor.py	Resizing, contrast and sharpness enhancement, and BGR→RGB conversion, shared by all entry points.
predictor.py	Wrapper around the LandingLens detection endpoint; returns predictions with confidence scores.
defect_typer.py	Generative diagnosis through the Mistral VLM; provides the fallback verdict for low-confidence cases, builds the prompt, encodes the image, parses the JSON report.
interpreter.py	Decision logic that converts raw predictions into a conform / non-conform verdict using the confidence threshold .
settings.py	Centralised configuration and secrets loaded from environment variables.
app.py	Streamlit dashboard orchestrating the full pipeline, history, charts, and e-mail alerting.

UML component view :



Part 2:

Abstract

In textile production, textile manufacturing relies heavily on visual quality control to detect defects surfaces in fabrics before other processes, while traditional manual inspection ways are inherently limited by human capacity, fatigue between operators. this affecting the industrial production had led us to this study that focuses on Fabric Defect Inspector.

this report presents our project QualityAgent, an agentic multi-tool system. the system operates in two-stage pipeline : a fine-tuned detection model (deployed through LandingLens) determines whether a fabric sample is defective and with what confidence score. This confidence dallas below 75% threshold, the decision is escalated to the Mistral VLM as a fallback and we take the higher confidence model results of both of them. When a defect is detect as the final decision, the Mistral VLM generates for us a structured diagnostic report containing all the necessary informations about te defect (defect type, severity, probable cause, recommended corrective action, a quality score and a final conformity verdict). the solution is delivered through a streamlit dashboard that supports both real-time webcam inspection and uploading an image, it also supports automatic e-mail altering and a history of all analysed defects.

Introduction

Thinking how could we combine both two paradigms and use their strongest aspects was the approach behind our project, QualityAgent, a system that precedes the CNNs(Convolutional Neural Networks) such as the YOLO family, reaches some limits with this area, since it demands substantial annotated Data to return only class labels, however the architecture of the Transformer reasoned deeply about images to generate Natural language descriptions in zero-shot way. So our project combines a supervised detector to filter incoming fabric images and a generative VLMthat both resolves low confidence cases and produce explainable diagnosis for the pieces flagged as defectuous. A great Working in real time and practical inspection tool.

State of the Art

Research on automated inspection has changed over time and we can break it down into four main groups of vision models.

Traditional (Handcrafted-Feature) Approaches

Traditional Defect Detection Approaches - Prior to the development of deep learning models for defect detection the defect detection process utilized two stages - a hand-crafted feature extraction stage, followed by a classification stage [Yurdakul, 2026]. The extracted features consisted of texture, edge, and gradient information and were extracted using a number of different descriptors e.g. GLCM, LBP, Gabor Filters, HOG, SIFT; and were classified classified using a number of different classifiers e.g. SVM, k-NN, Random Forest. The huge disadvantage of this method is the reliance on experts to manually create features which makes designing the feature-set an expensive and time-consuming endeavor: this deficiency

of traditional methods pushed the field toward developing automated feature learning techniques.[1]

Convolutional Neural Networks

Convolutional neural network applications include image segmentation (with the FCN, U-Net, and SegNet being popular examples). These models are built from convolution, pooling, and activation layers. One of the main problems is that deep networks can lose fine details due to pooling causing them to be unable to locate smaller fabric defects that occupy only a few pixels, as well as making it difficult for them to identify defects that appear infrequently in their training set. Ouyang et al. created a hybrid network with a customized activation layer using a defect probability map. As a result, the accuracy for counting defects is over 98%, precision is greater than 90% and the recall exceeds 80%. [2]

Vision Transformers

Transformers for Vision Transformers. Self-attention in Vision Transformers helps capture the long-range relationships throughout the entire image. This allows them to find defects missed by CNNs due to their small, limited local view. However, pure ViTs need significant computational resources and those resources must be used to train on a large number of samples because computing the attention of an image grows linearly with the size of the image (since it is processed as a complete simultaneous calculation). Hence, it is difficult to process real-time. To overcome this limitation, the recent trend has been to build lightweight hybrid models that use CNNs combined with transformers as their backbone. This new architecture exploits all of the advantages of using a CNN to extract features and applies those features through an efficient transformer. The hybrid models maintain both local texture information while also taking advantage of the global context, while continuing to be fast. These architectural approaches have now achieved greater than 99% F1 score on defect data sets of fabrics and added Grad-CAM explainability to the results, while using much fewer parameters than the larger transformer architectures.[3]

Vision Language Models

Vision Language Models (VLMs) are models that can combine information from various types of images and text in order to find defects in an image using natural language rather than traditional fixed class labels. One of the primary benefits of using VLMs for industrial inspection is their ability to be used for open vocabulary, zero-shot and few-shot detection by utilizing multimodal reasoning and being able to provide explanations for their predictions. This capability will be especially beneficial for fabric inspection, where the cost of acquiring defect samples is high and thus slows down production, making it highly advantageous to use models that can adapt and learn only from a small number of examples. Furthermore, VLMs can be combined with traditional computer vision techniques using lightweight fine-tuning to make VLMs appropriate for particular industrial applications. Recent efforts have demonstrated the effectiveness of adapting VLMs to detect surface defects using only a few training examples, however, performance varies across defect type and frequently produces false positive performance on more complex and irregular textured surfaces.[4][5]

Contribution:

Three main contributions associate with this project. A Two-Stage Architecture that separates the tasks of detecting defective items from diagnosing the defects. This permits the use of an efficient, supervised model detector to filter out all images. Once defective images are identified by the supervised fine-tuned model, then VLM model generates a unique diagnostic report for each defective image. The generative model additionally consulted whenever the supervised detector is not confident (the score is below 75%), acting as a verdict fallback before any typing. The generative score is structured and includes verdict fallback, defect type, severity level, probable cause, corrective action taken, quality of product, and a pass/fail status. The VLM is further operationalized as a public (zero-shot) diagnostic engine, so models can generate detailed diagnostic reports without any training on the specific defect present in the image. The architecture and integration of the JGF Framework into a real-time, traceable Tool that includes a dashboard display, a session history containing conformity statistics, and automatic e-mail alerts. This bridges the divide between the research prototype and a functional quality control system. A primary design consideration is cost and latency. Any given image processed through the Generative engine will incur a cost and a latency as the image transverses through an external API (Application Programming Interface). This would make it cost ineffective to process every image through the generative engine. By using a supervised model detector to identify only images with defects, and processing defective images only once every second through the generative engine, the Hybrid Architecture achieves a balance of responsiveness, accuracy and operating cost.

Results and Discussion

The supervised detection stage was run on an unseen held-out development data set consisting of 57 images that were not used to train the model. The average F1 score, precision, and recall were 98.2% during this test period. There were 25 true negative non-defective cases and an incorrect false positive, while the only other case (defective) was fully verified with 31 true positives and no false negatives — only one total sample error in the entire test period. We note that these results were obtained using the development data set and are not indicative of the model's performance on the training data because it performed 100% on its training data set; however, this would not be representative, as it had previously been trained on the same images.

The mAP@50 of the YOLOv11 trained on a larger collection of 9,354 images was an average precision of 79.9%, with an average precision of 80.7% and a mean recall of 76.5%. The YOLOv11 was able to localise the defect using a bounding box and classify the defect (for example, a hole with an 89% probability would be marked as a hole), which can be important information when determining the location of a defect.

Mistral's software created diagnoses across three categories of defect type that seem logical and have a degree of plausibility. Certain samples returned similar diagnoses in terms of defect type (i.e. relevant defect type), but there was a difference in how Mistral disclosed these. One example: consider a stained sample. The diagnosis returned indicated that stain was the defector, with a severity level of low, a quality score of 85/100 (very satisfactory), and that this was likely caused by a liquid contaminant during manufacturing, as well as recommending to treat the stained area. Conversely, for

the frayed item, the diagnosis returned indicated that a ‘loose thread’ was the defect; quality score – 90/100 (very satisfactory); probable cause – mechanical stress; recommendation – to trim off the frayed threads as per ISO standards.

Therefore, while two-stage processes provide good feedback through both categories of reference, the relationship between staged processes through supervised detectors versus VLM generators is critical to providing quality assurance, including recommendations for corrective actions (CA). The primary difference between these staging processes is in the way in which each system performs. The supervised detector is much more rapid, less expensive, and provides much greater accuracy at the binary decision level, it also covers the main detector’s uncertain results when the confidence score is under 75%, but does not reliably account for information and provides a non-descriptive feedback mechanism. On the other hand, the generative VLM is much slower, more expensive per call to the API, and relies on an external third party service (i.e., use of an external API). Nevertheless, by leveraging both approaches, QualityAgent can deliver a reliable detection and an explainable diagnosis. The most significant obstacle faced by the QualityAgent system at this time remains application of both approaches concurrently due to the limitations of using an external API to deliver the results from the VLM and the limited data (sample size) used for developing the currently deployed detector.

Conclusion

In conclusion, this paper presents a two-part, quality inspection system for fabrics called QualityAgent, which utilizes a finetuned supervised detector along with a zero-shot generative Vision Language Model. The supervised portion provides rapid and accurate fault detection, with a 98.2% F1 score on the validation set, and the generative portion produces structured, human-readable diagnostic reports. Integrated into a real-time dashboard with full traceability and automatic alerting, these components illustrate that traditional deep learning can be put together with modern generative AI to work as complementary components in an industrial quality control process. Future work will include adding a Retrieval-Augmented Generation layer on ISO quality standards, connecting the system to ERP/MES production systems, adding a fixed industrial camera on the production line, and enabling continuous re-training using operator feedback.

The present design addresses many recurring issues encountered in the prior analyses. First, while removing the annotation bottleneck limiting supervised detection by shifting the need for annotated examples from the binary detector to the VLM using zero-shot explanation generation. Second, it closes the explanation gap that results from most detectors and anomaly models providing only label or residual map outputs by converting a raw judgement into an actionable ISO-compliant report. Third, it evaluates the cost and latency considerations that make it infeasible to execute large generative models on each frame, as the inexpensive supervised gate is used to filter the incoming video stream, while the VLM is invoked approximately once per second and only on low confidence frames, where it acts as a fallback decision maker, and on confirmed defects, where it generates the diagnostic report. This balances responsiveness and accuracy while minimising operational costs. Finally, it reduces the research-to-production gap by providing a single deployable solution with all detection diagnosis/tracibility/conformance statistics/email alerting capabilities compared to an offline prototype solution.

Future work will focus on adding an RAG component for ISO quality standards, linking the system to ERP/MES manufacturing systems, mounting a permanent industrial camera within the production line, and enabling continuous retraining from operator feedback.

References

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